QuantumToolbox.jl: An efficient Julia framework for simulating open quantum systems

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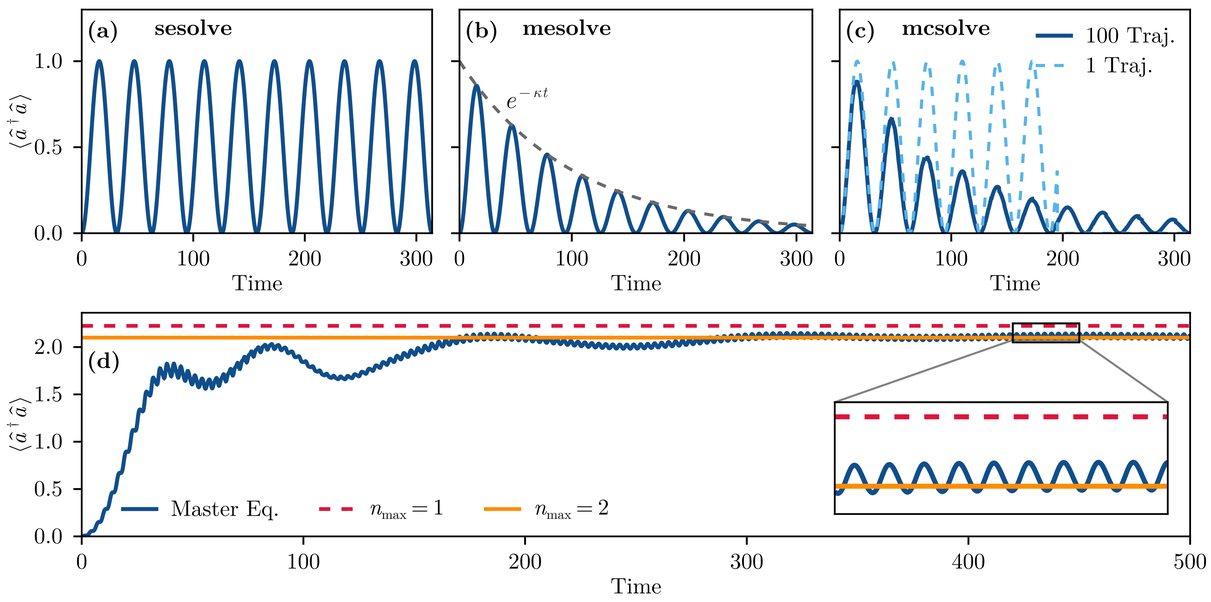
Motivation of the package

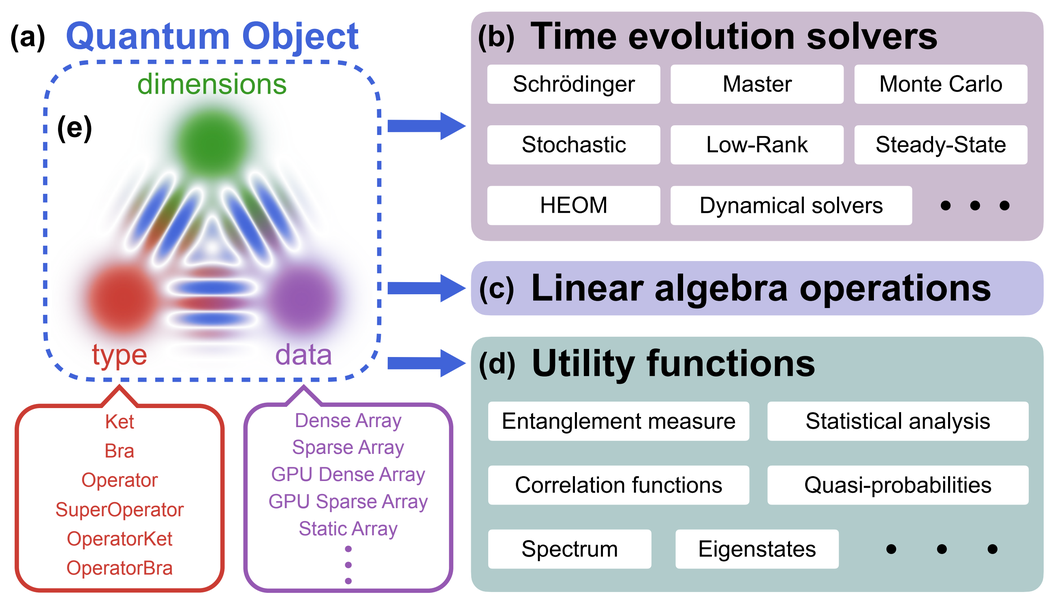


QuantumToolbox.jl provides an **efficient framework** for simulating open quantum systems in Julia. As an example we consider the Jaynes-Cummings model in presence of single photon and atomic losses. The density matrix evolves according to the **Lindblad master equation**

$$\frac{d\hat{\rho}}{dt} = -i[\hat{H},\hat{\rho}] + \kappa\mathcal{D}[\hat{a}]\hat{\rho} + \gamma\mathcal{D}[\hat{\sigma}_-]\hat{\rho} \quad \text{with} \quad \mathcal{D}[\hat{O}]\hat{\rho} = \hat{O}\hat{\rho}\hat{O}^\dagger - \frac{1}{2}\{\hat{O}^\dagger\hat{O},\hat{\rho}\}$$

```
ωc, ωa, g, κ, γ = 1, 1, 0.1, 0.01, 0.01

# cavity
N = 10 # Cutoff of the cavity Hilbert space
a = tensor(destroy(N), qeye(2)) #annihilation operator

# atom
σz = tensor(qeye(N), sigmaz()) # Pauli-Z operator
σm = tensor(qeye(N), sigmam()) # annihilation operator
σp = tensor(qeye(N), sigmap()) # creation operator

H = ωc * a' * a + ωa/2 * σz + g * (a * σp + a' * σm)

ψ0 = tensor(fock(N, 0), basis(2, 0))

C1 = sqrt(κ) * a
C2 = sqrt(γ) * σm
c_ops = [C1, C2]
e_ops = [a' * a] # average photon number

tlist = range(0, 10*π/g, 1000)
sol_me = mesolve(H, ψ0, tlist, c_ops, e_ops=e_ops)
```

Structure of the package

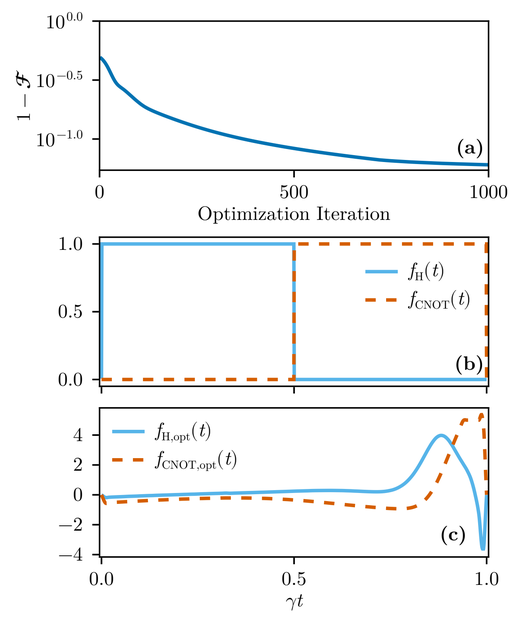


Automatic Differentiation

We consider a two-qubit system initialized in $|0,0\rangle$, with **the goal of generating the Bell state** $|\Phi^+\rangle = (|0,0\rangle + |1,1\rangle)/\sqrt{2}$ at a final time t_f. In the absence of losses, this state can be ideally prepared with the time-dependent Hamiltonian

$$\hat{H}_{\text{Bell}}(t) = \hat{H}_{\text{H}}^{(1)}(t) + \hat{H}_{\text{CNOT}}(t)$$

$$\hat{H}_{\text{H}}^{(1)}(t) = f_{\text{H}}(t)\frac{\pi}{t_f\sqrt{2}}(\hat{\sigma}_x^{(1)} - \hat{\sigma}_z^{(1)}) \quad \text{and} \quad \hat{H}_{\text{CNOT}}(t) = f_{\text{CNOT}}(t)\frac{\pi}{t_f}(\hat{I}^{(1)} + \hat{\sigma}_z^{(1)}) \otimes (\hat{I}^{(2)} - \hat{\sigma}_x^{(2)})$$

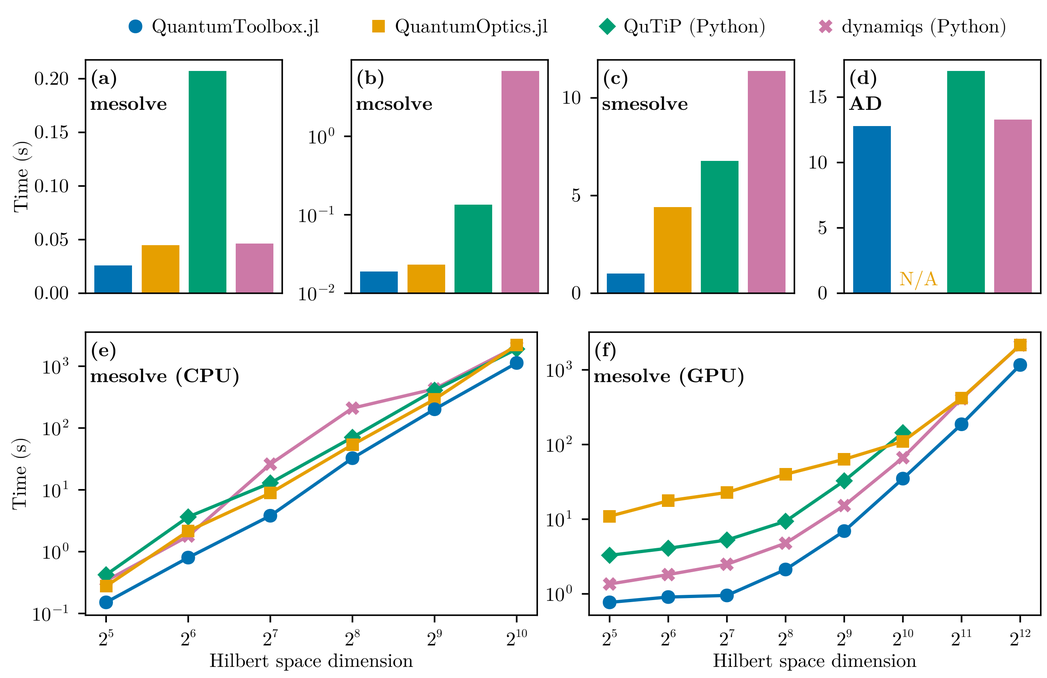


We now assume the functional forms of $f_{\text{H}}(t)$ and $f_{\text{CNOT}}(t)$ are unknown, and instead **optimize their shapes through gradient-based algorithms.**

$$\tilde{f}_{\text{H,opt}}(t) = \sum_{i=1}^{N} \theta_i^{\text{H}} \chi_i(t)$$

$$\tilde{f}_{\text{CNOT,opt}}(t) = \sum_{i=1}^{N} \theta_i^{\text{CNOT}} \chi_i(t)$$

Benchmarks



References

[1] A. Mercurio, et al., Quantum 9, 1866 (2025)

[2] J.R. Johansson, et al., Computer Physics Communications 183, 1760–1772 (2012)

[3] Sebastian Krämer, et al., Computer Physics Communications 227, 109 (2018)

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